Week 7: Linear Models
Machine Learning
CS1675
Fall 2023

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The response variable y is modeled as a linear combination of the input vector $\mathbf{x}$, parameterized by linear coefficients $\boldsymbol{\beta}$.

$$\mathbf{y}|\boldsymbol{\beta}, \mathbf{X}, \sigma \rightarrow \texttt{normal}(\mathbf{y}|\mathbf{X}\boldsymbol{\beta}, \sigma)$$

In other words, in this supervised learning task our prediction for a label $\hat{y}$ is equal to a linear combination of input features, e.g. $\hat{y} = \beta_0 + \beta_1 x$ if x has a single dimension.

The following equation represents the un-normalized posterior distribution. Note that we are using a constant, known σ, which is the standard deviation of the *data* and determines the spread of the likelihood function:

$$p(\boldsymbol{\beta}|\mathbf{y}, \mathbf{X}, \sigma) \propto \prod_{n=1}^{N} p(y_n|x_n, \boldsymbol{\beta}, \sigma)p(\boldsymbol{\beta})$$

In the remainder of this document, we will assume a diffuse, uninformative prior such that the posterior is proportional to the likelihood only. Because the prior is not informative, the MAP and MLE estimates of the optimal values of $\boldsymbol{\beta}$ will be identical.

1 Rewriting the log-likelihood with matrix math

The log-likelihood of the data is given by:

$$\log p(\mathbf{y}|\boldsymbol{\beta}, \mathbf{X}, \sigma) = -\frac{1}{2\sigma^2} \sum_{n=1}^{N} \left\{ \left(y_n - \sum_{d=0}^{D} x_{n,d}\beta_d \right)^2 \right\} \tag{1}$$

$$= -\frac{1}{2\sigma^2} \sum_{n=1}^{N} \left\{ (y_n - \mathbf{x_n}\boldsymbol{\beta})^2 \right\} \tag{2}$$

$\mathbf{X}$ is our design matrix, where the zeroth column is equal to 1 to account for the intercept, and each following column represents one dimension of our input $\mathbf{x}_n$. Thus, there are N rows representing each sample of $\mathbf{x}$, and $D+1$ columns for each dimension of the input plus the intercept.

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & x_{1,2} & \dots & x_{1,D} \\ 1 & x_{2,1} & x_{2,2} & \dots & x_{2,D} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{N,1} & x_{N,2} & \dots & x_{N,D} \end{bmatrix}$$

The nth sample of can be represented as a row vector:

$$\mathbf{x_n} = [1, x_1, x_2, \dots, x_D]$$

The model parameters which we are trying to learn are stored in a column vector:

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_D \end{bmatrix}$$

Equation (2) is simply a rewriting of (1) substituting the inner product for the summation symbol:

$$\sum_{d=0}^{D} x_{n,d}\beta_d = \mathbf{x_n}\boldsymbol{\beta}$$

We can further simplify Equation (2) with matrix multiplication. To demonstrate this, it is easiest to work backwards. In the lecture slides, we saw the following expression for the log-likelihood of our linear model (again, recall that we are modeling the input as a multivariate normal distribution).

$$\log p(\mathbf{y}|\boldsymbol{\beta}, \mathbf{X}, \sigma) = -\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Let us first expand the expression inside the parentheses, which represents the difference between the true response value and the prediction:

$$\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} y_1 - (\beta_0 + x_{1,1}\beta_1 + x_{1,2}\beta_2 + \cdots + x_{1,D}\beta_D) \\ y_2 - (\beta_0 + x_{2,1}\beta_1 + x_{2,2}\beta_2 + \cdots + x_{2,D}\beta_D) \\ \vdots \\ y_N - (\beta_0 + x_{N,1}\beta_1 + x_{N,2}\beta_2 + \cdots + x_{N,D}\beta_D) \end{bmatrix} \tag{3}$$

$$= \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_N \end{bmatrix} \tag{4}$$

Equation (4) is merely substituting the symbol ϵ ("epsilon") for the long equations in (3). This letter was chosen because the difference in these two vectors is the vector of *prediction errors*. Note that the result is a column vector, with each row representing the difference between predicted and observed values of the response given the set of parameters and training data.

Remember from the univariate case we want the exponent in our normal equation to contain the sum of squared errors. Two square the error vector according to matrix multiplication, it is necessary to take the inverse such that the dimensions match. Let $\mathbf{A}$ be a $m \times n$ matrix, and $\mathbf{B}$ be $n \times l$. Then the inner product $\mathbf{AB}$ will have dimension $m \times l$. If we multiply a row vector by a column vector, the result will be as scalar: the dot-product!

$$\begin{aligned} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) &= [\epsilon_1, \epsilon_2, \ldots, \epsilon_N] \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_N \end{bmatrix} \\ &= \epsilon_1 * \epsilon_1 + \epsilon_2 * \epsilon_2 + \cdots + \epsilon_N \epsilon_N \\ &= \sum_{n=1}^{N} \epsilon_n^2 \end{aligned}$$

2 Calculating the maximum likelihood estimate $\hat{\boldsymbol{\beta}}_{MLE}$

Next, we will find the MLE estimate of our parameters for the multivariate case. Recall that in the univariate case, we found the optimal value for our parameter by setting the derivative equal to zero and solving for the unknown. Now that we have multiple unknown parameters, we have to consider the gradient vector - a column vector with $D+1$ rows where each row is the derivative of the log-likelihood with respect to each $\beta_d \in 0, D$

$$\mathbf{g} = \frac{d\mathcal{L}}{d\boldsymbol{\beta}} = \begin{bmatrix} \frac{\partial L}{\partial \beta_0} \\ \frac{\partial L}{\partial \beta_1} \\ \frac{\partial L}{\partial \beta_2} \\ \vdots \\ \frac{\partial L}{\partial \beta_D} \end{bmatrix}$$

The following equation block calculates the derivative of the log-likelihood with respect to the intercept, β_0. You can check for yourself that the inner products in the final line are equivalent to the summations over all n samples.

$$\begin{aligned} \frac{\partial L}{\partial \beta_0} &= \frac{\partial}{\partial \beta_0}\left[-\frac{1}{2\sigma^2}\sum_{n=1}^{N}\left(y_n - \sum_{d=0}^{D} x_{n,d}\beta_d\right)^2\right] \\ &= -\frac{1}{2\sigma^2}\frac{\partial}{\partial \beta_0}\left[\sum_{n=1}^{N}\left(y_n - x_{n,0}\beta_0\right)^2\right] \\ &= -\frac{1}{2\sigma^2}\left[\sum_{n=1}^{N} -2x_{n,0}\left(y_n - x_{n,0}\beta_0\right)\right] \\ &= \frac{1}{\sigma}\left[\sum_{n=1}^{N} x_{n,0}y_n - \beta_0\sum_{n=1}^{N} x_{n,0}^2\right] \\ &= \frac{1}{\sigma}\left[\mathbf{x}_{:,0}^T\mathbf{y} - \mathbf{x}_{:,0}^T\mathbf{x}_{:,0}\beta_0\right] \end{aligned}$$

Next, let us calculate the MLE for the intercept only by setting its derivative equal to zero and solving for β_0. Recall that $\mathbf{x}_{:,0}$ refers to the first column of our design matrix, which is always equal to one. Our estimate for the best intercept is simply the mean of the response variable!

$$\begin{aligned} \hat{\beta}_0 \rightarrow \frac{1}{\sigma}\left[\mathbf{x}_{:,0}\mathbf{y} - \mathbf{x}_{:,0}^T\mathbf{x}_{:,0}\beta_0\right] &= 0 \\ N\bar{\mathbf{y}} - N\hat{\beta}_0 &= 0 \\ \hat{\beta}_0 &= \bar{\mathbf{y}} \end{aligned}$$

For the MLE estimates of other parameters, we have to actually consider the value of our input at each location. The final step has been written out in both matrix and sum notation:

$$\begin{aligned}
\hat{\beta}_1 \rightarrow \frac{1}{\sigma}\left[\mathbf{x}_{:,1}\mathbf{y} - \mathbf{x}_{:,1}^T\mathbf{x}_{:,1}\beta_1\right] &= 0 \\
\mathbf{x}_{:,1}^T\mathbf{y} - \mathbf{x}_{:,1}^T\mathbf{x}_{:,1}\hat{\beta}_1 &= 0 \\
\mathbf{x}_{:,1}^T\mathbf{y} &= \mathbf{x}_{:,1}^T\mathbf{x}_{:,1}\hat{\beta}_1 \\
\hat{\beta}_1 &= \mathbf{x}_{:,1}^T\mathbf{y}(\mathbf{x}_{:,1}^T\mathbf{x}_{:,1})^{-1} \\
&= \frac{\sum_{n=1}^N x_{n,1}y_n}{\sum_{n=1}^N x_{n,1}^2}
\end{aligned}$$

With some more matrix math, we can express the entire gradient vector in a single equation. Check the dimensions of each component to verify for yourself that the result is a $D+1 \times 1$ column vector.

$$\mathbf{g} = \frac{1}{\sigma}\left[\mathbf{X}^T\mathbf{y} - \mathbf{X}^T\mathbf{X}\boldsymbol{\beta}\right]$$

Thus our MLE estimates of the parameters are the set of $\boldsymbol{\beta}$ for which the gradient vector is equal to zero.

$$\hat{\boldsymbol{\beta}}_{MLE} = \boldsymbol{\beta}|\mathbf{g} = 0$$

Finally, we can write the Hessian matrix — the second-order derivatives of our log-likelihood — as the derivative of the gradient vector. Note that the value is a constant which depends only on the input data, not the learned parameters.

$$\mathbf{H} = \frac{d\mathbf{g}}{d\boldsymbol{\beta}} = -\frac{1}{\sigma}\mathbf{X}^T\mathbf{X}$$